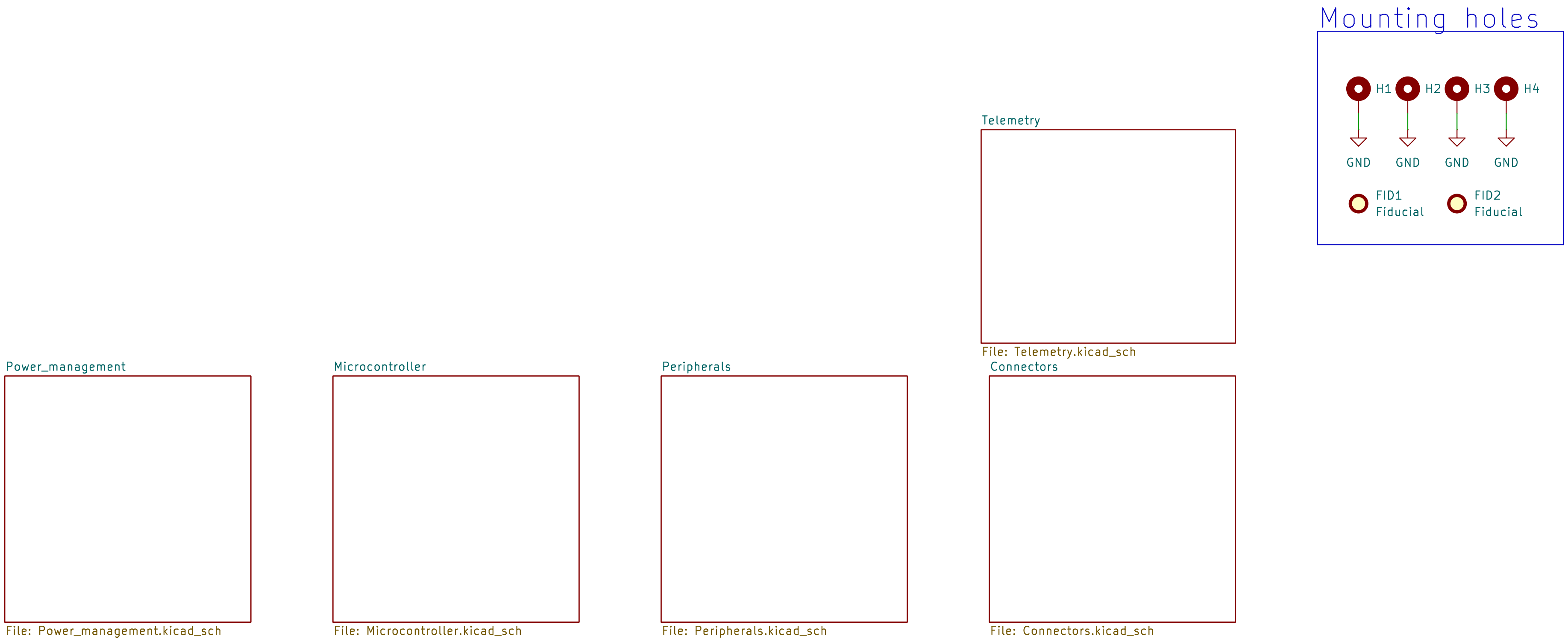

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Universal battery powered SDcard datalogger.

Author: Mlab www.mlab.cz

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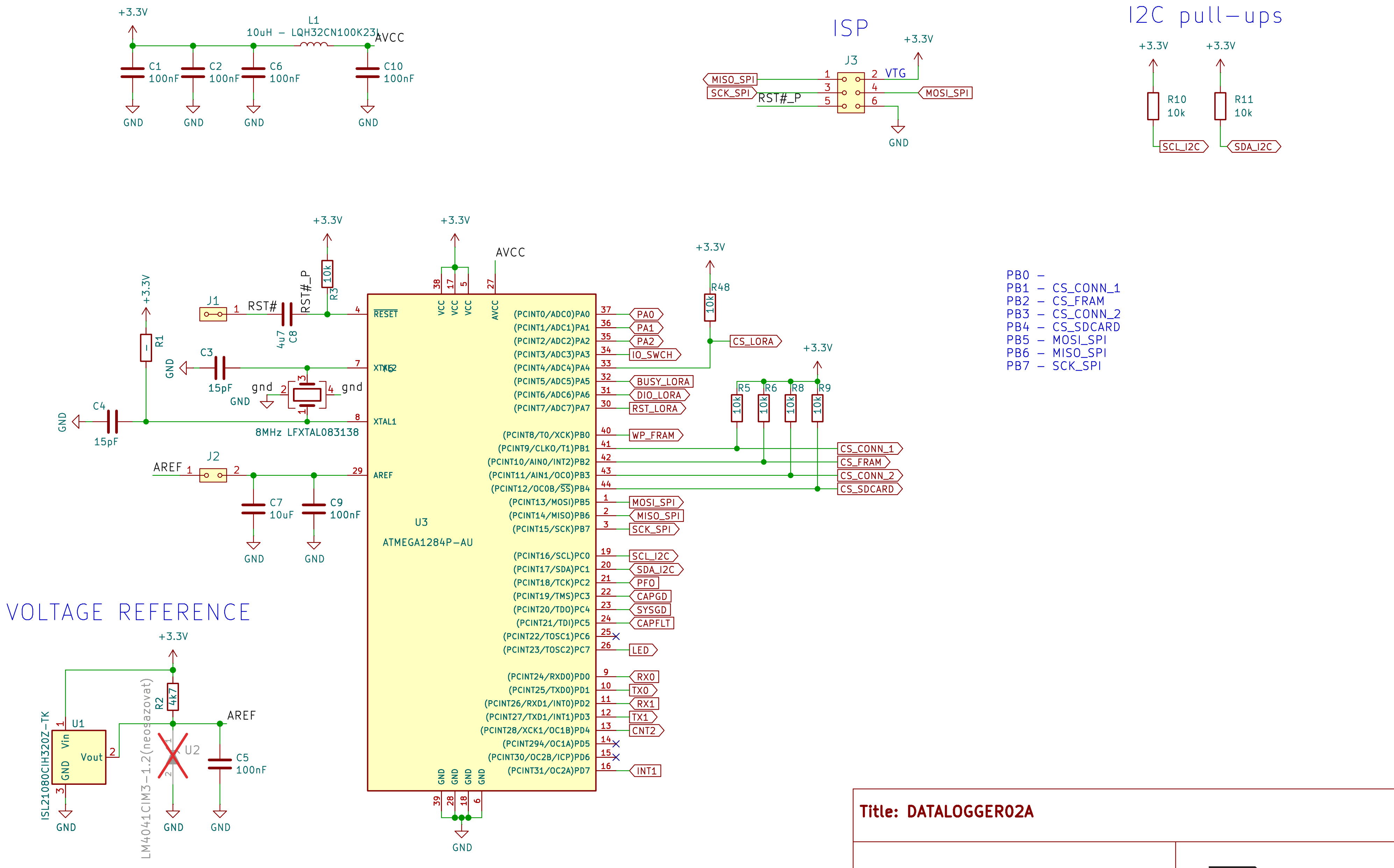

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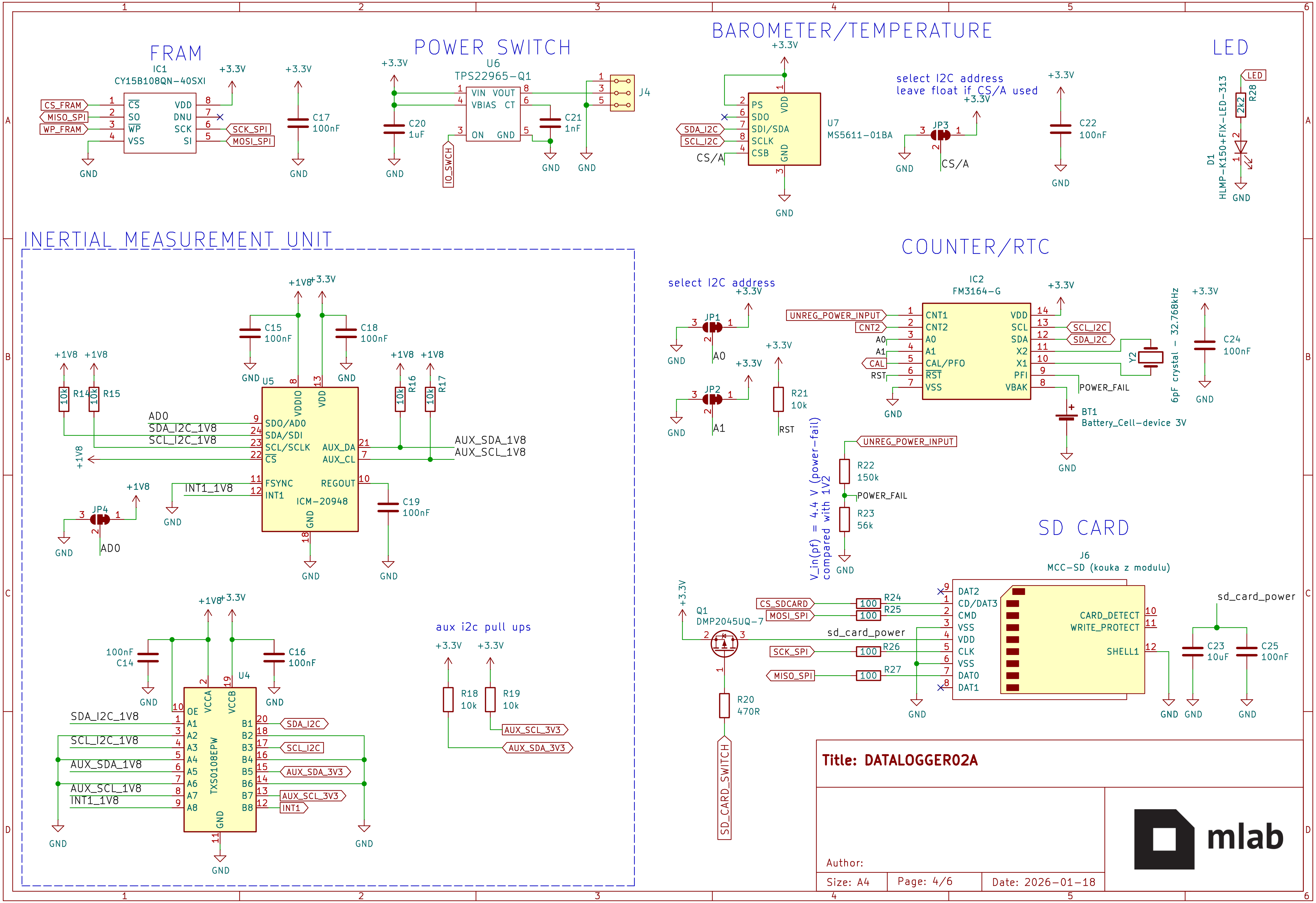

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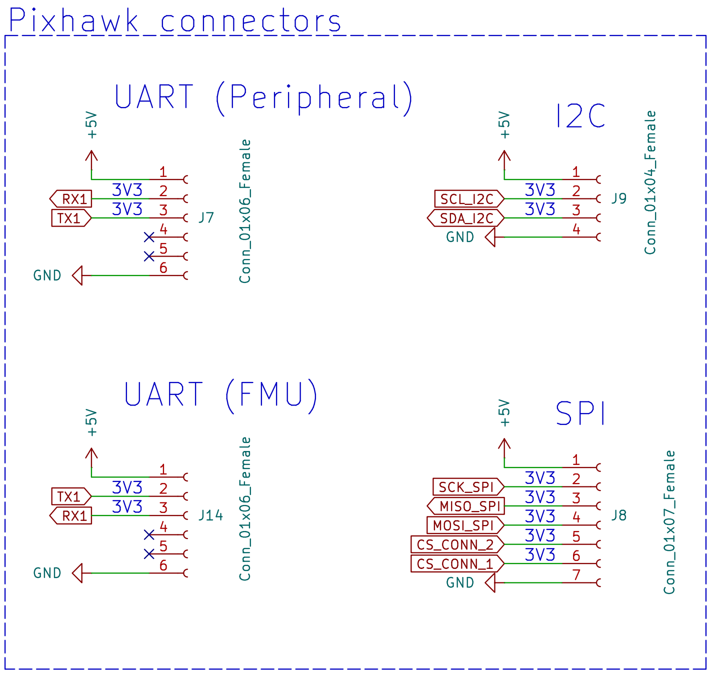

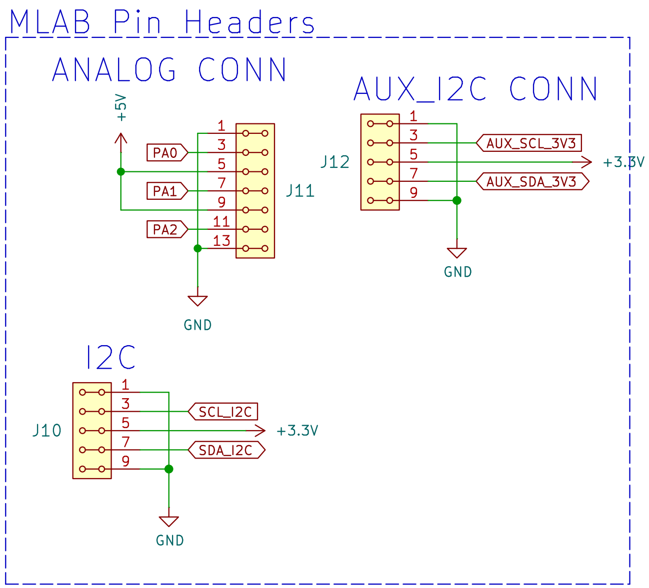

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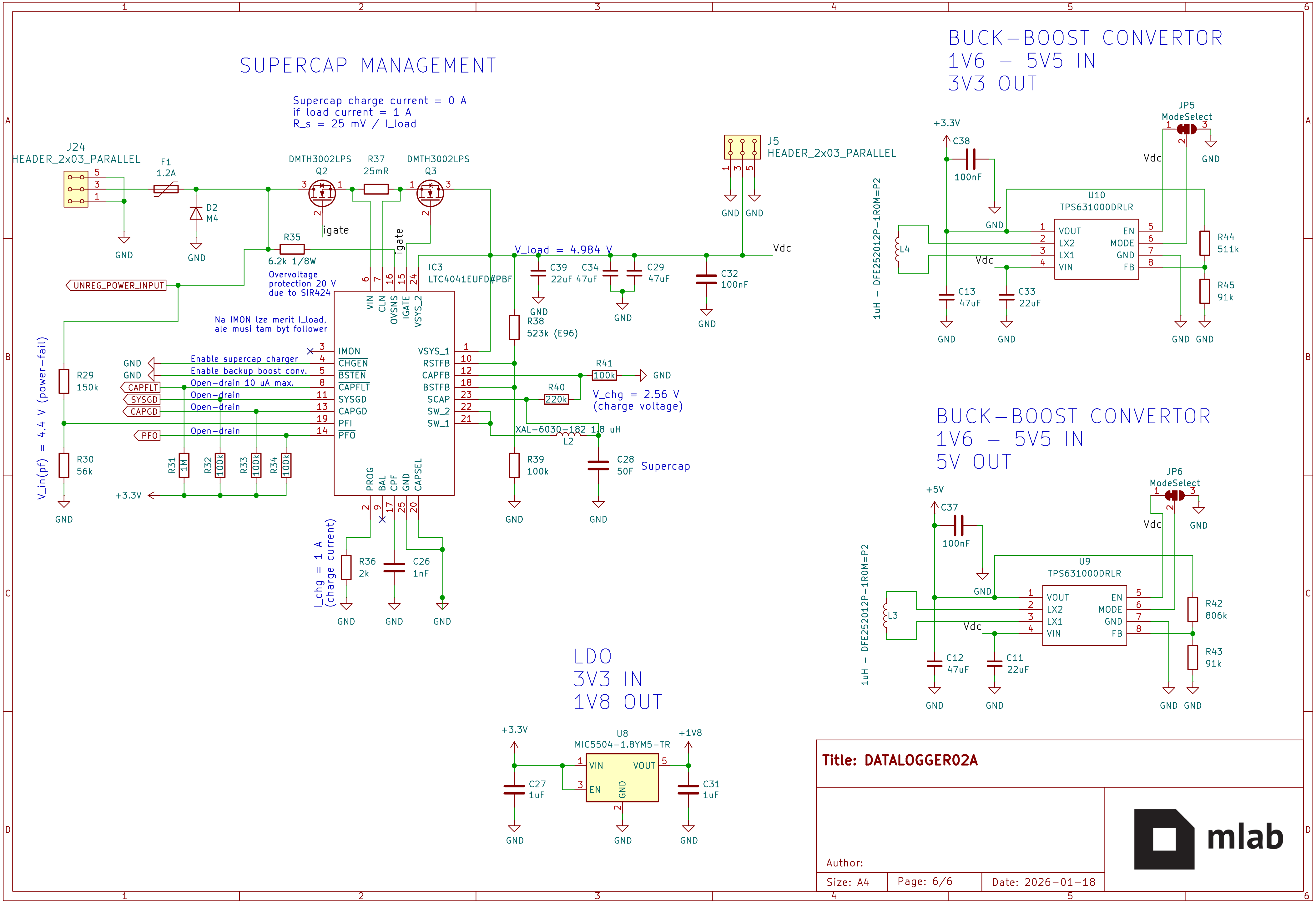
SUPERCAP MANAGEMENT
Supercap charge current = 0 A
if load current = 1 A
R_s = 25 mV / I_load
J24
HEADER_2x03_PARALLEL
F1
1.2A
D2
M4
DMTH3002LPS
Q2
R37
25mR
DMTH3002LPS
Q3
igate
R35
6.2k 1/8W
Overvoltage
protection 20 V
due to SIR424
UNREG_POWER_INPUT
IC3
LTC4041EUFD#PBF
V_load = 4.984 V
C39
22uF
C34
47uF
C29
47uF
C32
100nF
Vdc
J5
HEADER_2x03_PARALLEL
R38
523k (E96)
R41
100k
R40
220k
V_chg = 2.56 V
(charge voltage)
XAL-6030-182 1.8 uH
L2
C28
50F
Supercap
R39
100k
Na IMON lze merit I_load,
ale musi tam byt follower
Enable supercap charger
Enable backup boost conv.
Open-drain 10 uA max.
Open-drain
Open-drain
Open-drain
CAPFLT
SYSGD
CAPGD
PFO
R29
150k
R30
56k
V_in(pf) = 4.4 V (power-fail)
R31
1M
R32
100k
R33
100k
R34
100k
+3.3V
R36
2k
C26
1nF
I_chg = 1 A
(charge current)
GND
BUCK-BOOST CONVERTOR
1V6 - 5V5 IN
3V3 OUT
JP5
ModeSelect
C38
100nF
U10
TPS631000DRLR
L4
1uH - DFE252012P-1R0M=P2
C13
47uF
C33
22uF
R44
511k
R45
91k
BUCK-BOOST CONVERTOR
1V6 - 5V5 IN
5V OUT
+5V
JP6
ModeSelect
C37
100nF
U9
TPS631000DRLR
L3
1uH - DFE252012P-1R0M=P2
C12
47uF
C11
22uF
R42
806k
R43
91k
LDO
3V3 IN
1V8 OUT
U8
MIC5504-1.8YM5-TR
+1V8
C27
1uF
C31
1uF
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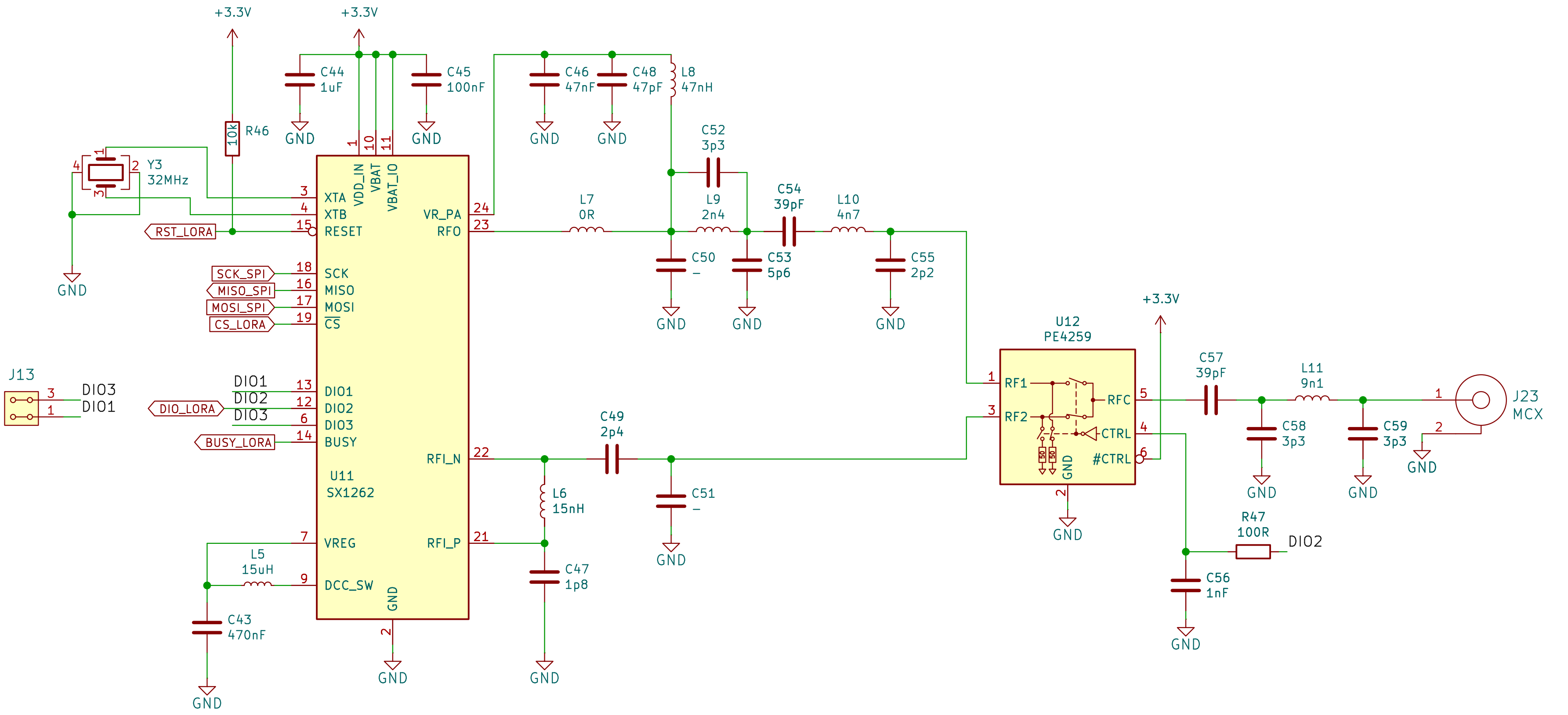

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